Shivendra Agrawal

Curriculum Vitae

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Research Interests: Human-Robot Interaction, Embodied AI, Vision-Language Models, Semantic Perception & Localization, Assistive Robotics, Human-Robot Co-Adaptation.

Academic Appointments

2026–Present **Postdoctoral Associate**, *Yale University*.
Interactive Machines Group; Advisor: Prof. Marynel Vázquez

Education

2019–2026 **University of Colorado, Boulder**, *Ph.D.*, Computer Science, *Advisor: Prof. Bradley Hayes*.

2017–2019 **University of Colorado, Boulder**, *MS*, Computer Science.

2009–2014 **Indian Institute of Technology, Kharagpur**, *Integrated BS + MS*, *Mathematics & Computing*.

Publications

In Submission/Development

In Submission **S. Agrawal**, B. Hayes. "VLM-GLoc: VLM-enhanced Mapping & Localization in geometrically and semantically aliased spaces." (Paper)

In Submission **S. Agrawal**, B. Hayes. "GIST: Multimodal Knowledge Extraction and Spatial Grounding via Intelligent Semantic Topology." (Website)

Refereed Conference/Journal Papers

RA-L 2025 **S. Agrawal**, A. Naik, J. Brawer, B. Hayes. "ShelfAware: Real-Time Visual-Inertial Semantic Localization in Quasi-Static Environments with Low-Cost Sensors." (Journal). (Website)

AAMAS 2023 **S. Agrawal**, S. Nayak, A. Naik, B. Hayes. "ShelfHelp: Empowering Humans to Perform Vision-Independent Manipulation Tasks with a Socially Assistive Robotic Cane." (Website)

IROS 2022 **S. Agrawal**, M. E. West, B. Hayes. "A novel perceptive robotic cane with haptic navigation for enabling vision-independent participation in the social dynamics of seat choice." (Website)

HRI 2019 **Best Paper Runner-up:** A. Tabrez, **S. Agrawal**, B. Hayes. "Explanation-based reward coaching to improve human performance via reinforcement learning." (Website)

Doctoral Consortium

AAMAS 2023 **S. Agrawal**. "Assistive robotics for empowering humans with visual impairments to independently perform day-to-day tasks" (Paper)

Workshops

ICRA 2026 **S. Agrawal**, A. Roncone, B. Hayes. "Distributional Semantics for Robust Global Localization in Cluttered, Geometrically Aliased Environments" (Paper)

IROS 2022 **S. Agrawal**, B. Hayes. "ShelfHelp: Empowering Humans to Perform Vision-Independent Manipulation Tasks with a Socially Assistive Robotic Cane." (Website)

Research Experience

2026–Present **Postdoctoral Associate**, *Interactive Machines Group*, Yale University.

- **Multi-Robot HRI in Shared Workspaces:** Studying how people perceive and interact with robots in busy, high-stakes everyday environments.

2019–2026 **Graduate Research Assistant**, *Collaborative AI and Robotics Lab*, University of Colorado Boulder.

- **Active Semantic Localization – Current Work:** Large indoor spaces such as hospitals and colleges have sparse features making global localization challenging. This work focuses on acquiring relevant information for robust global localization.
- **Generative AI & Spatial Grounding (GIST):** A multimodal pipeline that transforms a point cloud into a lightweight semantic topology, enhancing spatial grounding for VLMs and enabling intent-aware search, global localization, and visually grounded language-based navigation.
- **Semantic Localization (ShelfAware):** Designed a *Semantic Particle Filter* that utilizes quasi-static landmarks as mathematical distributions for robust localization. Integrated RGB-D and Visual-Inertial Odometry (VIO) to achieve state-of-the-art accuracy on low-cost hardware.
- **Learning Grasp Guidance Policies from Human Demonstrations (ShelfHelp):** Created a robotic cane system that provides online grasping guidance. Implemented a Markov Decision Process planner learned from human demonstrations to optimize verbal instruction timing.
- **Social Goal-Finding:** Developed a goal-finding framework modeling privacy and intimacy norms and integrated it with path-planning and haptics-based guidance. Conducted human-subject studies validating increased confidence in reaching the goal.
- **Explainable Robotic Coaching:** Developed a second-order "Theory of Mind" framework allowing robots to infer human mental models and provide targeted, explainable coaching by observing suboptimal actions (Best Paper Runner-up, HRI '19).
- **Industry-Sponsored Collaboration (NEC Corporation):** Built a proof-of-concept system using a Vision-Language Model (VLM) and a mobile robot to autonomously detect and explain spatial anomalies. (Video Demo)

Teaching Experience

Instructor CSCI 3104 Algorithms - *Best Part-Time Graduate Instructor* (2019-2020).

TA CSCI 3104 Algorithms - *Outstanding TA award* (2018, 2019).

CSCI 3002 Fundamentals of Human Computer Interaction.

CSCI 3302 Intro to Robotics - *Outstanding TA award* (2022).

CSCI 4302/5302 Advanced Robotics (2023).

CSCI 5322 Algorithmic Human-Robot Interaction (2025).

CSCI 3202 Intro to AI (2026).

Industry Experience

Sep 2016 – May 2017 **Associate Data Scientist**, Noodle.ai.

- **Automated Analytics:** Engineered automated pipelines for Exploratory Data Analysis (EDA) to evaluate large-scale datasets, reducing manual analysis time by 40%.
- **Predictive Modeling:** Designed and deployed predictive models for demand forecasting and churn analysis for major enterprise clients.

Mar 2016 – Aug 2016 **Co-Founder**, Progyrus.

- **System Architecture:** Architected a distributed task management system for large-scale vernacular translation, managing work allocation for 50+ concurrent translators.
- **Full-Stack Development:** Designed RESTful APIs (Django) and the frontend interface (AngularJS) for the order management platform.

2014 – 2016 **Senior Engineer (Analytics)**, Robert Bosch GmbH.

- **Optimization Algorithms:** Developed a clique-based algorithm to optimize vehicle prototype configurations, satisfying complex boolean constraints to reduce physical testing costs.
- **NLP:** Engineered a Python-based GUI (Tkinter) for automated log analysis and ticket clustering, utilizing NLP techniques to identify recurring vehicle faults.

Service & Mentorship

Organizer	Idea Mixer Seminar Series, NSF STC for Human and Robot Co-Adaptation (2026–Present)
Peer Review	Reviewer for CoRL, ACM/IEEE HRI, IEEE ICRA, IEEE IROS, ACM THRI
Leadership	Social Media Chair (HRI 2024), Volunteer (AAMAS 2023)
Mentorship	Mentored 5 M.S. and 3 Undergraduate researchers; mentees placed at Tesla, Meta, and AWS.

Talks and Guest Lectures

Jul 2026 **Ph.D. Thesis Defense: Foundations for Context-Aware Assistive Guidance**, *University of Colorado Boulder*, Boulder, CO, (Recording).
Integrating implicit semantic, social, and physical cues for assistive guidance.

Jun 2026 **Research Talk: Context-Aware Embodied AI Systems for Human-Centered Environments: From Assistive Guidance to Autonomous Robots**, *Yale University*, New Haven, CT.

Jun 2026 **Invited Talk: Context-Aware Embodied AI Systems for Human-Centered Environments**, *National Institute of Standards and Technology (NIST)*.

Feb 2026 **Guest Lecture: Hidden Markov Models**, *Intro to AI*, Boulder, CO, (Slides).
Guest lecture for the Intro to Artificial Intelligence (CSCI 3202) class.

Apr 2024 **Robotics Seminar Talk: Context-Aware Robotics**, *University of Colorado Boulder*.
Presented thesis work on context-aware assistive guidance to the broader CU Boulder audience.

Apr 2024 **Guest Lecture: Practical Computer Vision**, *Franklin & Marshall College*.
Guest lecture for Intro to Machine Learning (CPS 360).

Sep 2023 **Leading with Impact alumni event**, Denver, CO, (Event news).
Panel discussion on the current state and future of AI.

Grants, Fellowships, and Awards

2026	Dissertation Completion Fellowship.
2026	Selected as the Next Gen Cohort for CERAWeek.
2022–2026	Best Poster, Annual Research Expo, CU Boulder (2022, 2023, 2026).
2024	Beverly Sears Graduate Student Grant (competitive research proposal, $1,000), CU Boulder.
2024	Bell Family Endowed CS Scholarship.
2024	Publication Recognition Award, CU Boulder.
2023	AAMAS Doctoral Consortium.
2022, 2023	Conference Support Fellowship, CU Boulder.
2021-2022	David T. Spalding Graduate Teaching Fund Fellowship.
2019-2020	Best Part-Time Graduate Instructor Award.
2017-2019	Outstanding TA Awards.

Technical Skills

LLM / VLM	Foundation Models (Qwen, Gemma), Local Deployment, PyTorch, LangGraph, Claude Code
Robotics	ROS, SLAM, Particle Filters, WeBots, RViz, Sensor Fusion, OptiTrack, LiDAR
Languages	Python (NumPy, SciPy), C++, JavaScript, R
Methodologies	Human-Subject Experiment Design, Statistical Analysis